# NEXONIX REFLEXO PVT. LTD.

DPIIT & MSME Approved

CAD Designing Internship 2026

# INTERNSHIP REPORT

on

## Design, Assembly and Engineering Documentation of a Belt-Support Roller Unit Using SolidWorks


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Internship Duration: 30/08/2026 to [End Date], 2026

Academic Year 2026

# Certificate

[ Paste the Internship Completion Certificate issued by
Nexonix Reflexo Pvt. Ltd. here ]

# Student Declaration

I, **[Student Name]**, a student of [Course / Branch] at [Name of Institute / University], hereby declare that the work presented in this internship report, titled *"Design, Assembly and Engineering Documentation of a Belt-Support Roller Unit Using SolidWorks"*, is original and was carried out by me during the CAD Designing Internship 2026 at Nexonix Reflexo Pvt. Ltd.

All parts, assemblies, drawings and screenshots included in this report were created by me while completing the internship assignments. Where ideas, standards, books or online documentation have been consulted, they have been acknowledged in the References chapter. This report has not been submitted, in whole or in part, to any other institute or organisation for the award of any degree, diploma or certificate.

I understand that, as stated in the internship guidelines, the assignments are individual work, and that collaboration on ideas is permitted but copying of files or screenshots is not.

Place: _____

Date: _____

Signature: _____

([Student Name])

# Acknowledgement

I would like to express my sincere gratitude to **Nexonix Reflexo Pvt. Ltd.** for giving me the opportunity to be part of the CAD Designing Internship 2026. The structured programme of lectures and graded assignments gave me a clear path from basic sketching to complete engineering documentation.

I am especially thankful to my company mentor and instructor, **Mr. Anmol Verma**, for his clear explanations, step-by-step assignment briefs and prompt guidance throughout the four weeks. His emphasis on clean FeatureManager trees, fully defined sketches and proper documentation has shaped the way I approach design work.

I thank my faculty mentor, Mr. Deep Singh Thakur, and the Head of the Department, Mr. Saurabh Tripathi, at Rajkiya Engineering College, Banda for permitting me to undertake this internship and for their constant encouragement and academic support.

I also thank my fellow interns for the healthy exchange of ideas during the programme, and my family and friends for their patience and motivation while I completed the assignments and this report.

**ARPIT MISHRA**

# Abstract

This report presents the work carried out during the four-week CAD Designing Internship 2026 conducted by Nexonix Reflexo Pvt. Ltd. under the guidance of instructor Anmol Verma. The internship was organised around seven lectures and four graded assignments, and its objective was to build practical, industry-relevant competence in computer-aided design using SolidWorks Premium 2022 SP1.0.

In Week 1 the focus was on sketching fundamentals: three basic profiles (a 60 x 40 mm rectangle, a Ø50/Ø20 mm ring and an L-shape) and an independent mounting-bracket sketch with base plate, upright arm, two Ø10 mm holes and perpendicular and coincident constraints. In Week 2 the work moved to part modelling and advanced features. A symmetrical mounting bracket was produced using Extrude, Mirror, Flex and Rib, and a thin-walled triangular phone stand was produced using Circular Sketch Pattern, Draft, Shell and Fillet. In Week 3 five components, namely a roller wheel, two flanged bushes, a shaft, two pillow-block supports and a base plate, were modelled individually and brought together in a seven-component assembly using mates, after which a production-style A3 engineering drawing with orthographic and isometric views and dimensions was generated. Week 4 was devoted to consolidation, verification of the FeatureManager trees, preparation of exports and screenshots, and the preparation of this report.

The selected project for detailed discussion is the **Belt-Support Roller Unit**, a compact roller carried on a fixed shaft between two supports on a common base plate. The report describes its objectives, scope and applications, the hardware and software used, the testing and verification carried out, the results obtained, the challenges faced and the improvements that could follow, such as material assignment, motion study, tolerancing and finite-element validation.

The internship strengthened my skills in feature-based parametric modelling, assembly mating, drafting, dimensioning and documentation, and also developed soft skills such as time management, communication through GitHub and LinkedIn, and systematic problem solving.

**Keywords:** CAD, SolidWorks, Part Modelling, Assembly, Mates, Engineering Drawing, Mirror, Shell, Rib, Flex.

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# Chapter 1: Internship Objectives

The CAD Designing Internship 2026 at Nexonix Reflexo Pvt. Ltd. was designed to take a beginner from the first sketch on a workplane to a fully constrained assembly and a production-style drawing. The overall aim was to be able to model real, manufacturable components, to assemble them logically, and to communicate the design through standard documentation. This chapter explains the objectives that were achieved, with examples from the assignments completed.

## 1.1 Technical Skills

The primary objective was to acquire working proficiency in a parametric 3D CAD package. The following technical objectives were achieved:

- **Sketching:** drawing precise 2D profiles, applying dimensions and geometric relations such as perpendicular, coincident and fixed, and bringing sketches to the Fully Defined state (Assignment 1).
- **Part modelling:** creating solids with Extruded Boss/Base, Extruded Cut, Revolved Boss/Base, Chamfer and Fillet. For example, the roller wheel, bush and shaft were produced as revolved bodies, while the base plate and support were built from extrusions.
- **Advanced features:** using Mirror, Flex, Rib, Shell, Draft and reference geometry to produce the symmetrical mounting bracket and the phone stand (Assignment 2).
- **Assembly design:** inserting components, fixing a base component and applying mates so that parts relate to one another as they would in a real mechanism (Assignment 3).
- **Engineering drawing:** generating orthographic and isometric views on an A3 sheet and adding dimensions, centre marks and annotations using the Annotation toolbar.

## 1.2 Practical Learning

Every lecture was paired with a hands-on assignment, so each concept was applied immediately. Practical learning came from following a defined sequence of operations, observing the FeatureManager design tree grow feature by feature, and correcting errors when a feature failed or a sketch was under-defined. The requirement that screenshots show the full FeatureManager tree together with the model taught me to treat the tree as a record of design intent rather than as a by-product of modelling. Exporting the work as .SLDPRT, .SLDASM, .SLDDRW, .STL or .STEP files also showed how CAD data moves between design, manufacturing and other software.

## 1.3 Team Collaboration

Although every assignment was individual work, the internship encouraged collaboration on ideas. Doubts were discussed with the instructor and with fellow interns, approaches to features such as Flex trim planes or the circular sketch pattern were compared, and the final deliverables were shared in a way that others could review: source files and screenshots were published to a public GitHub repository and the screenshots were posted on LinkedIn, with the links submitted through the internship portal. This workflow resembles the way design teams share and review data in industry, and it taught me to organise files and name them clearly so that others can follow the work.

## 1.4 Problem Solving

Several tasks required analytical thinking rather than simple repetition. Examples include deciding which plane to use as the mirror plane so that both bolt holes of the bracket remained symmetric; choosing the correct trim-plane distances and triad location so that the Flex feature bent the plate through 90 degrees; trimming the offset triangle correctly before extruding the phone stand's back support with a 24 degree draft; and determining which mate types would let the wheel spin on its shaft while the supports and bushes stayed positioned. In each case the approach was to isolate the problem, test one change at a time, and use the error messages and the status bar (for example Under Defined) as feedback.

# Chapter 2: Weekly Internship Activities

The internship ran for four weeks. Each week combined lectures with one graded assignment, and each assignment carried a fixed due date. Table 1 summarises the programme, and the sections that follow describe each week in terms of learning topics, assignments, screenshots, skills learned and outcomes.

Table 1: Summary of weekly programme

| Week | Lectures covered | Assignment | Due |
|---|---|---|---|
| 1 | Lecture 1: Introduction to CAD Designing<br>Lecture 2: Sketching Fundamentals | Assignment 1: Sketch It Right (2D Profile and Bracket Challenge) | Day 7 |
| 2 | Lecture 3: Part Modelling Basics<br>Lecture 4: Advanced Part Modelling | Assignment 2: Build It Better (Advanced Part Modelling Challenge) | Day 14 |
| 3 | Lecture 5: Assembly Design<br>Lecture 6: Engineering Drawings<br>Lecture 7: Surface Modelling | Assignment 3: Build It Complete (Assembly, Drawing and Surface Design) | Day 21 |
| 4 | Consolidation and documentation | Assignment 4: Internship Report | End of internship |

## 2.1 Week 1: Sketching Fundamentals

**Learning topics.** Week 1 introduced the CAD working environment, planes, the origin and the sketch workflow. The key ideas were sketch entities, dimensions, geometric constraints and sketch status. A central habit taught in this week was to anchor the sketch to the origin first, so that the sketch never floats and can reach the Fully Defined (all-black) state.

**Assignment 1, Part A: three basic profiles.** A 60 x 40 mm rectangle, a ring with a Ø50 mm outer diameter and a Ø20 mm inner hole, and an L-shape made from an 80 x 20 mm base and a 20 x 50 mm upright were created as separate shapes. The ring required concentric placement of the two circles and the L-shape required the two rectangles to be joined exactly at a corner.

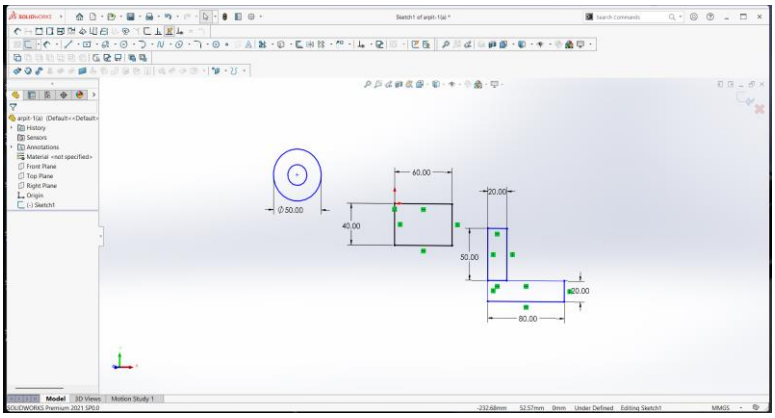


Figure 1: Rectangle profile (60 x 40 mm) with its FeatureManager tree

Ring profile (Ø50 mm outer, Ø20 mm inner hole)

L-shape profile built from two joined boxes (80 x 20 mm base and 20 x 50 mm upright)

**Assignment 1, Part B: mounting bracket sketch.** The mounting bracket from the Lecture 2 activity was recreated without step-by-step instructions. It consists of an 80 x 30 mm base plate, a 20 x 50 mm upright arm starting at the top-left corner of the base, a Ø10 mm hole on the base centred 20 mm from the left edge, and a Ø10 mm hole on the arm centred 25 mm from the arm's bottom. The constraints applied were Perpendicular between base and arm, Coincident at the shared corner, and a fixed corner on the origin. In the screenshot below, the model also shows the holes produced by the two Cut-Extrude features.

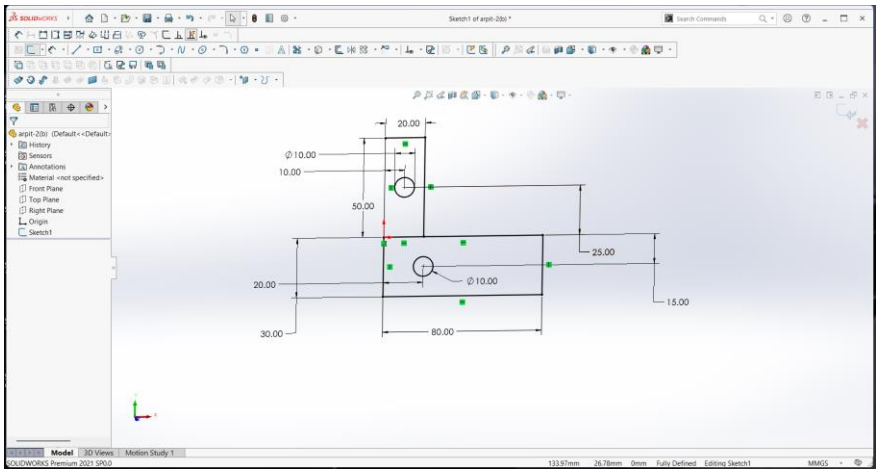



Figure 4: Mounting bracket sketch model (pg1D) with base, upright arm and two Ø10 mm holes;

Table 2: Assignment 1 evaluation scheme (as given in the assignment brief)

| Item | Part | Marks |
|---|---|---|
| Rectangle profile (60 x 40 mm) with dimensions visible | A | 10 |
| Ring profile (Ø50 outer, Ø20 hole) | A | 10 |
| L-shape profile (80 x 20 base + 20 x 50 upright) | A | 10 |
| Bracket sketch with all four components at correct dimensions | B | 25 |
| Constraints applied correctly (perpendicular + coincident) | B | 25 |
| Sketch status: Fully Defined | B | 20 |
| **Total** | | **100** |

**Skills learned and outcome.** By the end of Week 1 I could create dimensioned, constrained sketches, convert them into simple solids and read the sketch status. The outcome was a set of four clean models with exports and screenshots prepared for GitHub submission.

## 2.2 Week 2: Part Modelling and Advanced Features

**Learning topics.** Lectures 3 and 4 covered part-modelling basics and advanced features: Linear Pattern, Mirror, Shell, Draft, Fillet, Rib and Flex. These features capture design intent such as symmetry, wall thickness and manufacturability.

**Assignment 2, Task 1: symmetrical mounting bracket.** A 100 x 40 mm rectangle was sketched on the Top Plane and extruded to 5 mm. A Ø6 mm hole was cut near one corner (15 mm from the left edge and 10 mm from the top edge) and then mirrored about the Right Plane and Front Plane to obtain a symmetric hole pattern. The Flex feature, with a bend angle of –90 degrees, trim planes at 45 mm and the triad positioned as specified, bent the plate into an L-shaped bracket. A reference plane offset by 10 mm carried the sketch for a support rib of 3 mm thickness with a 1 degree draft, which was then mirrored about the Front Plane.

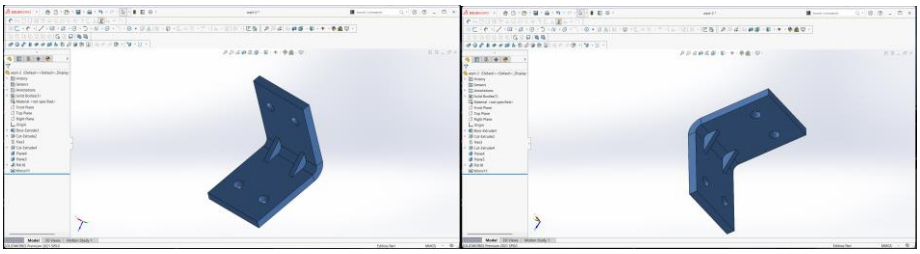


Figure 5: Symmetrical mounting bracket (2A); tree: Boss-Extrude1, Cut-Extrude1, Mirror1, Flex1, Plane1, Rib1, Mirror2

**Assignment 2, Task 2: phone stand.** A 120 mm equilateral triangle with 5 mm corner fillets was extruded 3 mm downward from the Top Plane, keeping the origin at the centre. A 30 x 5 mm slot, positioned 2 mm from the bottom edge, was repeated three times around the origin using Circular Sketch Pattern and extruded 6 mm upward to form phone-holding slots. A smaller triangle, obtained by converting the triangle's edges, offsetting them inward by 17 mm and trimming, was extruded 40 mm with a 24 degree draft to form the back support. The body was then shelled to 2 mm with the bottom face removed, and the edges were rounded with a 0.5 mm fillet.

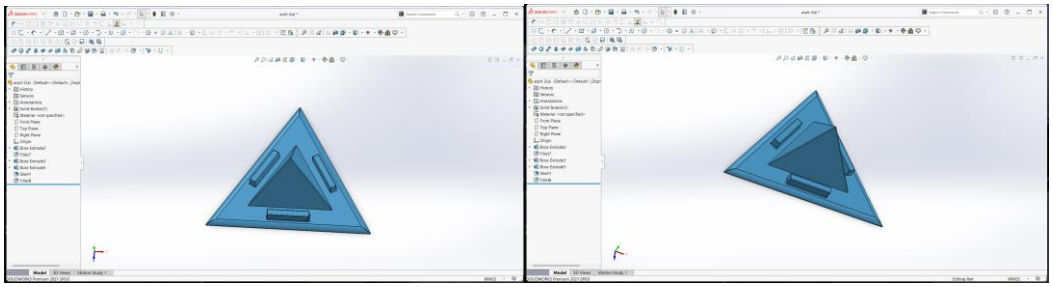


Figure 6: Phone stand (2B); tree: Boss-Extrude2, Fillet2, Boss-Extrude3, Boss-Extrude4, Shell1, Fillet3

Table 3: Features used in Assignment 2

| Component | Key features | Design purpose |
|---|---|---|
| Mounting bracket (2A) | Boss-Extrude, Cut-Extrude, Mirror, Flex, Reference Plane, Rib | Symmetry, 90-degree bend, stiffening |
| Phone stand (2B) | Boss-Extrude, Circular Sketch Pattern, Draft, Shell, Fillet | Phone slots, back support, light weight, safe edges |

**Skills learned and outcome.** I learned to use patterns and mirrors to avoid repeating work, to use Flex to deform a flat body, to add ribs and drafts for strength and moulding, and to use Shell and Fillet to produce thin-walled, manufacturing-ready

parts. The two parts were exported and their screenshots, showing the full tree and the model, were submitted.

## 2.3 Week 3: Assembly, Drawing and Surface Design

**Learning topics.** Lecture 5 introduced assembly design (inserting components, fixing the first component, mates and degrees of freedom), Lecture 6 covered engineering drawings (orthographic views, dimensions and basic GD&T) and Lecture 7 introduced surface modelling. Assignment 3 asked for a multi-part assembly (Task 1) and a complete engineering drawing of that assembly (Task 2).

**Assignment 3, Task 1: multi-part assembly.** Each part was first modelled individually and then brought into an assembly file. The five unique parts are shown in Figures 7 to 11.

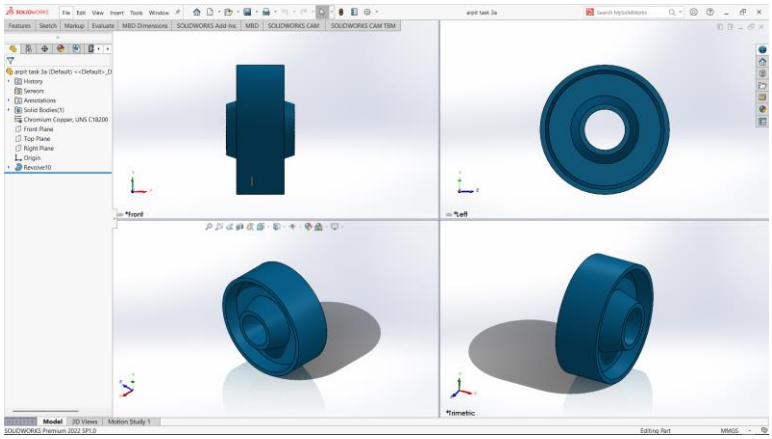


Figure 7: Roller wheel (Part1), a single Revolve feature

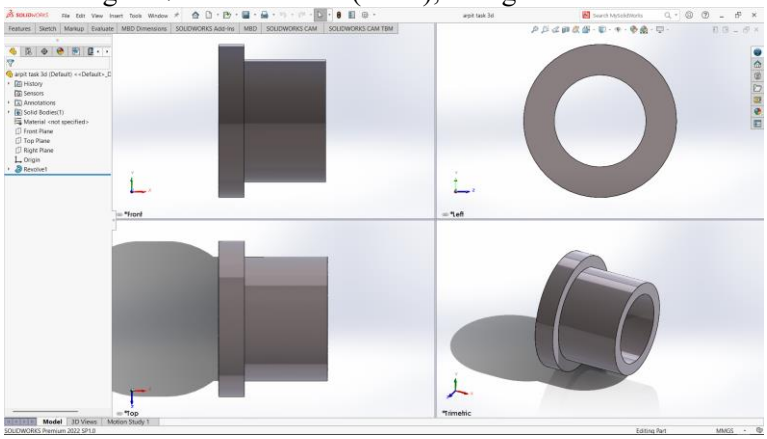


Figure 8: Flanged bush (Part2), Revolve and Cut-Extrude features

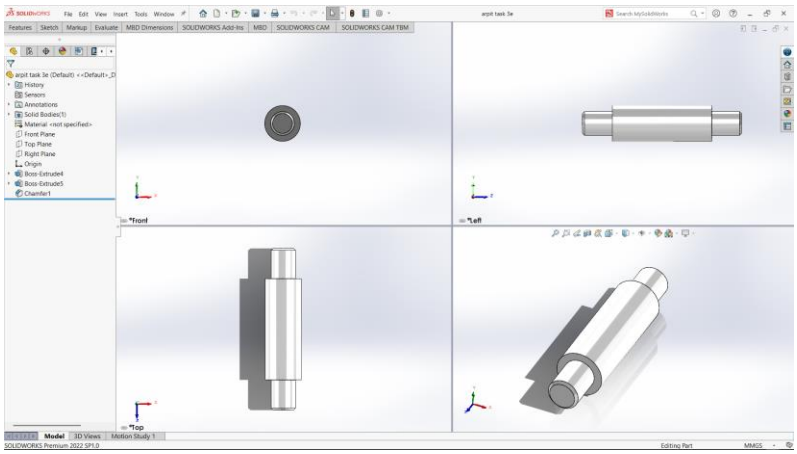


Figure 9: Stepped shaft (Part3), Revolve and Chamfer

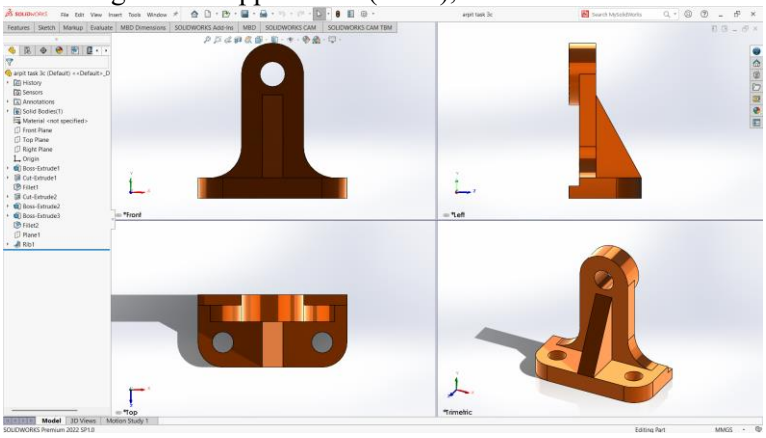


Figure 10: Pillow-block support (Part4): four Boss-Extrude features, a reference plane and a Rib

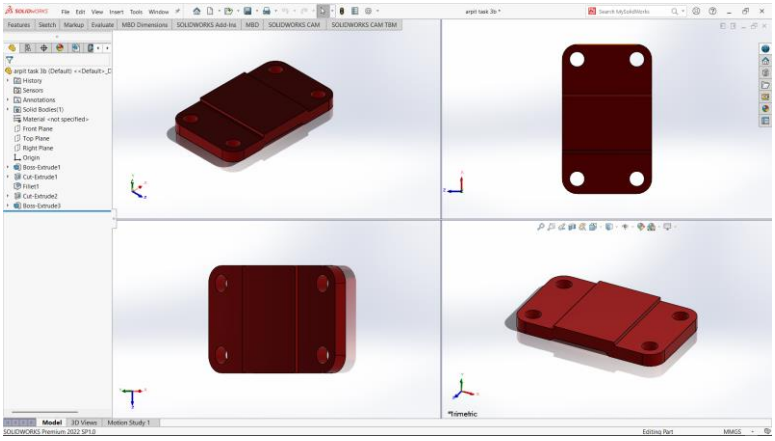


Figure 11: Base plate (Part5): two Boss-Extrude features and a Cut-Extrude for the four mounting holes

**Assembly.** The base plate was set as the fixed component, shown with the (f) prefix in the tree. The remaining components, namely the supports, the bushes, the shaft and the wheel, were added and positioned with mates, which are listed in the Mates folder of the assembly tree. Two copies of the support (AM3D) and two copies of the bush (AM3B) were used, because the wheel is carried on both sides.

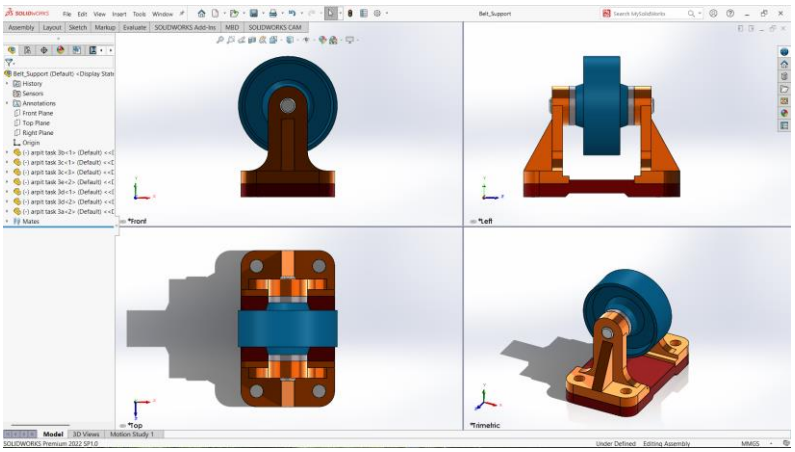


Figure 12: Assembly (Assem1) with seven components in four views and the Mates folder in the tree

**Assignment 3, Task 2: engineering drawing.** A new drawing on an A3 sheet was created from the assembly. It contains a top view, a front view, a side view and an isometric view, with overall dimensions, hole diameters, radii and centre marks added using the Annotation toolbar.

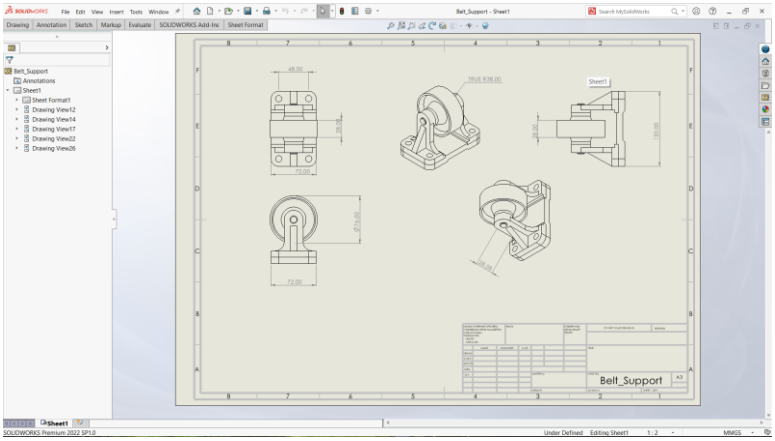



Figure 13: Engineering drawing of the assembly (AMAssem1) on an A3 sheet showing the Annotation toolbar

**Skills learned and outcome.** I learned how mates remove degrees of freedom, why the first component is fixed, how an assembly status such as Under Defined is interpreted, and how to move from a 3D model to a dimensioned 2D drawing. The outcome was a seven-component assembly and a production-style drawing, ready for export as .SLDASM and .SLDDRW.

## 2.4 Week 4: Consolidation, Documentation and Report

**Learning topics.** The final week was used to consolidate all work from the first three weeks into a presentable portfolio and to document it. This followed the submission guidelines repeated in every assignment brief: the FeatureManager tree must be clean with no red flags, screenshots must show both the full tree and the model, exports must be uploaded to GitHub and screenshots shared on LinkedIn.

**Activities.** Every part, assembly and drawing was reopened and rebuilt to confirm that there were no rebuild errors. Sketches were checked for their Fully Defined status. The files were exported in the required formats (.SLDPRT, .SLDASM, .SLDDRW, .STL or .STEP), organised by assignment, and published with their screenshots. This report (Assignment 4) was then prepared following the Nexonix Reflexo internship report template: front matter, seven chapters, references and the formatting guidelines.

**Skills learned and outcome.** The week improved my skills in technical documentation, version control and file organisation. The outcome was a complete, reviewable record of the internship and this report.

# Chapter 3: Project Introduction

For detailed discussion, the project selected from the internship work is the **Belt-Support Roller Unit**, which is the multi-part assembly of Assignment 3 together with its engineering drawing. It was selected because it uses almost every skill from the internship: sketching, extrusion, revolution, ribs, chamfers, patterns, mates and drafting. The unit consists of a roller wheel turning on a stepped shaft, which is carried through two flanged bushes in two upright pillow-block supports, with both supports bolted to a common base plate.

## 3.1 Objectives

The objectives of the project were:

- to model each component individually as a manufacturable part with a clean FeatureManager tree;
- to assemble the parts so that the wheel can rotate while the supports, bushes, shaft and base plate remain correctly positioned, with the base plate as the fixed component;
- to ensure that mating features, such as the shaft diameter, bush bore, support hole and wheel bore, are geometrically compatible;
- to produce a complete, dimensioned A3 engineering drawing with orthographic and isometric views;
- to document the model with screenshots and exports suitable for review and sharing.

## 3.2 Scope

The scope of the project covers the 3D modelling of five unique parts (seven components in the assembly), the assembly with mates, and the drawing of the assembly in SolidWorks Premium 2022 SP1.0. Dimensions are in millimetres (MMGS unit system). The project is a geometric and documentation exercise. Material properties, tolerances and fits, stress analysis and manufacturing process planning are outside the scope of the work completed during the internship and are noted as future improvements in Chapter 5.

Table 4: Bill of materials of the Belt-Support Roller Unit (item numbers follow the Assignment 3 drawing)

| Item | Component | Part file | Qty | Main features |
|---|---|---|---|---|
| 1 | Base plate | Part5 (AM3E) | 1 | Boss-Extrude (x2), Cut-Extrude (four mounting holes) |
| 2 | Pillow-block support | Part4 (AM3D) | 2 | Boss-Extrude (x4), Plane, Rib |
| 3 | Flanged bush | Part2 (AM3B) | 2 | Revolve, Cut-Extrude |
| 4 | Roller wheel | Part1 (AM3A) | 1 | Revolve |
| 5 | Stepped shaft | Part3 (AM3C) | 1 | Revolve, Chamfer |

The assembly status in the screenshot shows the base plate (AM3E) marked (f), meaning it is fixed in space, while the other components are marked (-), meaning they are free to move until constrained by mates.

Table 5: Key overall dimensions read from the assembly drawing

| Feature | Dimension (mm) |
|---|---|
| Overall length of base plate (top view) | 120 |
| Overall height of the assembly (front view) | 110 |
| Width of the support (side view) | 72 |
| Wheel outer diameter (side view) | Ø70 |
| Mounting hole diameter | Ø12 |

## 3.3 Applications

A roller carried on a fixed shaft between two supports is a basic building block in many machines. Typical applications are:

- belt and conveyor supports, where the roller guides or supports a moving belt;
- idler and tensioner rollers in small transmission and material-handling systems;
- guide rollers for sheet, film or cable feeding;
- caster-style or trolley wheels carried on a bracket;
- training and demonstration models for teaching assembly, mating and drafting in engineering courses.

The same modelling approach, a base component with mirrored supports and a rotating member, also applies to other products such as pulley blocks, bearing housings and hinge assemblies.

# Chapter 4: Hardware / Software Used

This chapter lists the hardware and software that supported the internship work. The software details are taken from the working environment visible in the screenshots, in which the application banner reads SOLIDWORKS Premium 2022 SP1.0 and the unit system is shown as MMGS.

## 4.1 Hardware

Table 6: Hardware used

| Item | Specification | Purpose |
|---|---|---|
| Computer | [Laptop / desktop model] | Running CAD software and preparing the report |
| Processor | [e.g. processor name and speed] | Rebuilding features and assemblies |
| Memory (RAM) | [e.g. 8 GB / 16 GB] | Handling multi-part assemblies and drawings |
| Graphics | [Integrated / dedicated GPU] | Smooth 3D rotation and shaded views |
| Display | [Screen size and resolution] | Viewing four-view layouts and the A3 drawing |
| Input devices | Keyboard and mouse | Sketching, selecting and dimensioning |
| Internet connection | Broadband / mobile data | Attending lectures, GitHub and LinkedIn submission |

## 4.2 Software

Table 7: Software used

| Software | Version / type | Purpose |
|---|---|---|
| SolidWorks | Premium 2022 SP1.0 | Part, assembly and drawing modelling |
| Tinkercad | Web application (optional for Assignment 1 Part A) | Basic shape profiles using workplane and ruler tools |
| GitHub | Web platform | Publishing exports and screenshots |
| LinkedIn | Web platform | Sharing screenshots as required by the assignments |
| Word processor / PDF viewer | Any | Preparing and reviewing this report |

## 4.3 Programming Languages

The internship was based on feature-based parametric modelling, so no programming language was required. The models are created through sketches, features and mates rather than through code. Only basic Git operations or the GitHub web interface were used to upload files.

## 4.4 Libraries

No external code libraries were used. Within SolidWorks, the built-in A3 sheet format and the standard drawing-view and annotation tools were used for documentation. Reference geometry, such as planes and the origin, was used as the base for features.

## 4.5 Tools

Table 8: SolidWorks tools used in the internship

| Tool group | Tools | Used in |
|---|---|---|
| Sketch and dimension | Sketch entities, Smart Dimension, relations, Circular Sketch Pattern, Convert and Offset Entities, Trim | All assignments |
| Base features | Extruded Boss/Base, Revolved Boss/Base | Assignments 1, 2 and 3 |
| Cut features | Extruded Cut, Revolved Cut | Brackets, bush and base plate |
| Pattern and symmetry | Mirror | Mounting bracket |
| Form and detail | Flex, Rib, Draft, Shell, Fillet, Chamfer | Brackets, phone stand, support and shaft |
| Reference geometry | Plane | Bracket rib and support |
| Assembly | Insert Components, Mate, Fixed component | Assembly (Assignment 3) |
| Drawing and annotation | Model views, Smart Dimension, Center Mark, Centerline, Note, Tables | Assembly drawing |

# Chapter 5: Results and Discussion

This chapter presents how the models were tested, the outputs obtained, an analysis of the modelling effort, the challenges faced, and possible improvements.

## 5.1 Testing

In CAD work, testing means verifying that the model is error-free, correctly constrained and consistent with the specification. The following checks were carried out for the parts, assembly and drawing, and are recorded in Table 9.

Table 9: Verification checks and observations

| Check | How it was verified | Observation |
|---|---|---|
| Rebuild errors | FeatureManager tree inspected after the last feature | No error or warning icons in any part tree (Figures 1 to 13) |
| Sketch status | Sketch colour and status bar checked | Sketches dimensioned and constrained to the origin |
| Symmetry | Mirrored holes and ribs viewed in Top and Left views | Holes and ribs of bracket 2A are symmetric |
| Thin-wall check | Shell1 and Fillet3 viewed in the tree of phone stand 2B | Shell and fillet applied without failure |
| Assembly state | Assembly status bar and tree prefixes | Base plate fixed (f); status bar reads Under Defined, consistent with the wheel retaining rotational freedom |
| Component fit | Front, Left and Top views of assembly | Wheel sits between the supports; shaft and bushes are aligned |
| Drawing | Views, dimensions and title block inspected | Four views on A3 sheet with dimensions and centre marks |

## 5.2 Output

The outputs of the internship are the dimensioned models of the bracket and phone stand, the assembled roller unit, and its drawing. Figure 14 shows the shaded views of the two Week 2 parts, and Figure 15 shows the view of the finished assembly.

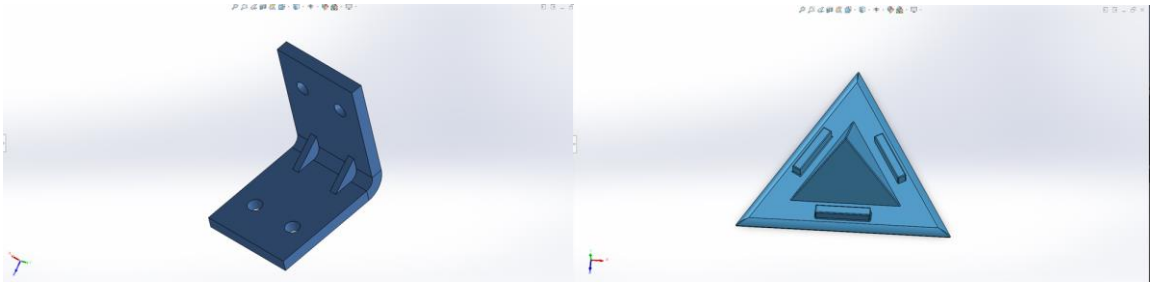


Figure 14: Views of the mounting bracket (left) and the phone stand (right)

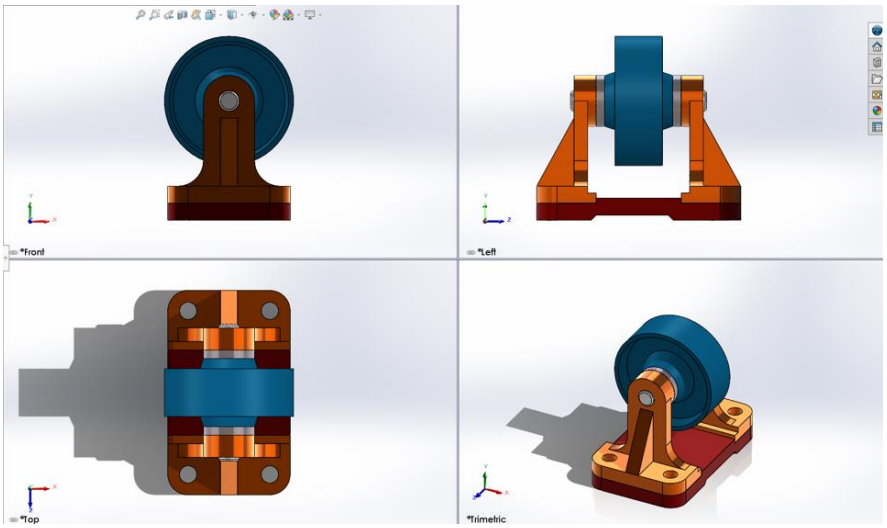



Figure 15: Views of the Belt-Support Roller Unit assembly

The engineering drawing in Figure 16 presents the assembly in four views, with overall dimensions of 120 mm in length and 110 mm in height, a 72 mm support width, a Ø70 mm wheel and Ø12 mm holes, along with radii and centre marks.

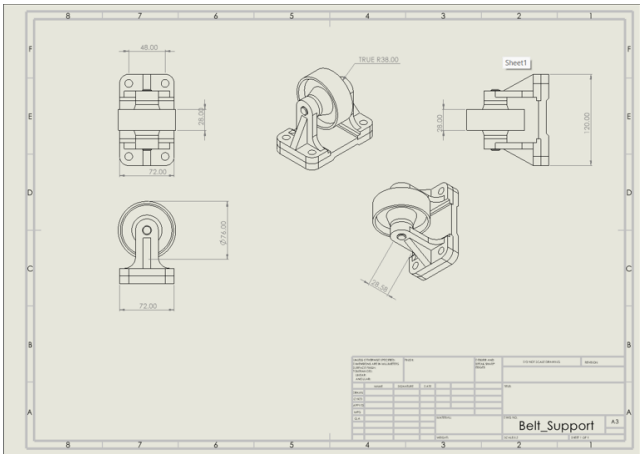



Figure 16: Engineering drawing sheet of the assembly (A3) with views and dimensions

## 5.3 Graphs

Figure 17 compares the number of features used in each model, taken from the FeatureManager trees in the screenshots. The count rises from one feature for simple profiles in Week 1 to seven features for the mounting bracket in Week 2, which reflects the increasing use of mirror, flex and rib to capture design intent. In Week 3 the counts depend on the nature of the part: revolved parts such as the wheel (1 feature) and the shaft (2 features) need few features, while the support (6 features) needs several because it combines a base, a lug, a boss and a rib.

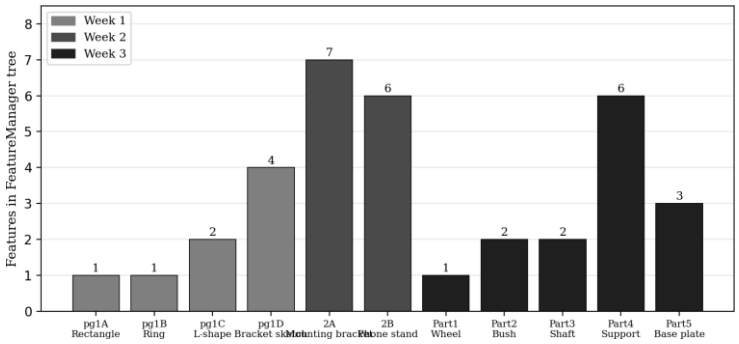



Figure 17: Number of features in the FeatureManager tree of each model

## 5.4 Challenges

The main challenges encountered, and how they were addressed, were:

- **Flex feature setup.** The bend depends on the bend angle, the two trim planes and the triad position and angle together. Small errors gave an incorrect bend, so the values were entered exactly as specified and checked in the Front view.
- **Selecting the correct planes for mirroring.** The mirror result depended on choosing the right plane for the model's orientation, so the plane was verified before confirming the feature.
- **Draft and offset sketch.** The 24 degree draft on the back support of the phone stand only works when the offset triangle is trimmed cleanly. Untrimmed corners caused sketch errors.
- **Compatibility between parts.** The shaft, bushes, supports and wheel had to share compatible diameters and lengths. This was handled by modelling to the drawing dimensions and checking the assembly views.
- **Mating order.** Applying mates in an order that kept the assembly stable and avoided conflicting mates required planning, with the base plate fixed first.
- **Drawing clarity.** With four views in one sheet, dimensions had to be placed so that they did not overlap, which required adjusting views and dimension positions.

## 5.5 Future Improvements

The project can be extended in several ways:

- **Material assignment.** The tree shows the material as not specified. Assigning materials such as steel for the shaft and a polymer for the wheel would give mass properties.
- **Motion study.** A motion study could animate the rolling wheel and confirm the clearance between the wheel and supports.

- **Fully defined configurations.** Maintaining one configuration with zero degrees of freedom and another with a moving wheel, as the brief describes, would document both locked and working states.
- **Tolerances and GD&T.** Adding fits for the shaft and bush and geometric tolerances to the drawing, in line with ASME Y14.5 [3], would make the drawing production-ready.
- **Bill of materials and balloons.** Adding balloons and a BOM table to the drawing and an exploded view to the assembly.
- **Analysis.** A static study on the support and base plate would check strength and could guide further use of ribs.
- **Surface-modelled handle.** The Week 3 overview also covers surface design; a surface-modelled ergonomic handle that mates to the assembly would extend the project.
- **Manufacturing.** The phone stand could be 3D printed from its STL export to validate wall thickness and draft.

# Chapter 6: Skills Learned

This chapter summarises the skills developed during the internship.

## 6.1 Technical Skills

The internship developed working skill in parametric CAD: sketching with constraints, creating extruded and revolved solids, applying mirror, flex, rib, draft, shell, fillet and chamfer, assembling components with mates, and creating dimensioned drawings. I also learned to read a drawing and reproduce a part from it, as in the Assignment 3 assembly, where the drawing was the specification. An important technical habit was keeping the design tree clean and logical.

## 6.2 Soft Skills

Soft skills included time management, since each assignment had a strict due date and late submissions were not accepted; attention to detail, since a small dimensional error affects later features; and self-discipline in following specifications. Working from written assignment briefs also improved my ability to interpret instructions and ask precise questions.

## 6.3 Communication

Communication skills improved through presenting work visually. Screenshots had to show the full tree and model, GitHub repositories had to be organised, and LinkedIn posts had to describe the work clearly. Writing this report improved my technical writing, including the use of captions, tables and references.

## 6.4 Leadership

Leadership during the internship took the form of ownership. I planned the sequence of each assignment, set my own milestones before the due dates, took responsibility for the quality of my submissions, and helped peers by discussing approaches without sharing files, in line with the academic-integrity rule. This is the foundation of leading a design task in a team.

## 6.5 Industrial Exposure

The internship gave exposure to industrial practice in several ways: parts designed with manufacturability in mind (draft angles, shelling, fillets), assembly practices (fixed base component, mates), standard drawing sheets (A3 with a title block), neutral file formats (.STEP and .STL) and the use of version control and professional networking platforms. It also showed that a design is complete only when it is documented well enough for someone else to build it.

# Chapter 7: Conclusion

## 7.1 Summary

The CAD Designing Internship 2026 at Nexonix Reflexo Pvt. Ltd. gave a complete, progressive introduction to computer-aided design. Starting with simple dimensioned profiles in Week 1, the work grew to a symmetrical mounting bracket and a thin-walled phone stand in Week 2, and then to a seven-component Belt-Support Roller Unit with a full engineering drawing in Week 3. Week 4 consolidated all of this into a documented portfolio and this report.

The Belt-Support Roller Unit demonstrated how individual parts, each modelled with an appropriate set of features, can be assembled with mates around a fixed base component and then communicated through a drawing. The verification checks showed clean feature trees and a logically mated assembly, while also identifying clear next steps: materials, motion study, tolerancing, analysis and a surface-modelled handle.

## 7.2 Learning Outcomes

On completing the internship, I am able to:

- create fully dimensioned and constrained 2D sketches and convert them into 3D solids;
- apply extrude, revolve, mirror, flex, rib, draft, shell, fillet and chamfer features appropriately to meet a design requirement;
- build a multi-part assembly using a fixed base component and mates, and interpret the assembly status;
- generate orthographic and isometric drawing views on a standard sheet and add dimensions and annotations;
- organise, export and share CAD data with screenshots through GitHub and LinkedIn;

- prepare a structured technical report with figures, tables and references.

This experience has prepared me to take on further CAD work, and I am grateful to Nexonix Reflexo Pvt. Ltd. for the opportunity.

# Chapter 8: References

References are listed below in IEEE format, which is used for in-text numbered citations such as [1] and [3]. The same sources are then shown in APA format for comparison and grouped by type.

## 8.1 IEEE Format


[1] D. C. Planchard, *Engineering Design with SOLIDWORKS 2022*. Mission, KS, USA: SDC Publications, 2022.

[2] N. D. Bhatt and V. M. Panchal, *Engineering Drawing: Plane and Solid Geometry*. Anand, India: Charotar Publishing House.

[3] *Dimensioning and Tolerancing*, ASME Standard Y14.5-2018, American Society of Mechanical Engineers, New York, NY, USA, 2019.

[4] R. G. Budynas and J. K. Nisbett, *Shigley's Mechanical Engineering Design*, 11th ed. New York, NY, USA: McGraw-Hill, 2020.

[5] G. Boothroyd, P. Dewhurst, and W. Knight, *Product Design for Manufacture and Assembly*, 3rd ed. Boca Raton, FL, USA: CRC Press, 2010.

[6] J. J. Shah and M. Mäntylä, *Parametric and Feature-Based CAD/CAM: Concepts, Techniques, and Applications*. New York, NY, USA: Wiley, 1995.

[7] A. A. G. Requicha, "Representations for rigid solids: Theory, methods, and systems," *ACM Comput. Surveys*, vol. 12, no. 4, pp. 437–464, Dec. 1980.

[8] Dassault Systèmes, "SOLIDWORKS Help." help.solidworks.com (accessed [Month Day, 2026]).

[9] Autodesk, "Tinkercad." tinkercad.com (accessed [Month Day, 2026]).

[10] Nexonix Reflexo Pvt. Ltd., "CAD Designing Internship 2026: Assignments 1 to 4," internship course material, 2026.


## 8.2 APA Format


Planchard, D. C. (2022). *Engineering design with SOLIDWORKS 2022*. SDC Publications.

Budynas, R. G., & Nisbett, J. K. (2020). *Shigley's mechanical engineering design* (11th ed.). McGraw-Hill.

Requicha, A. A. G. (1980). Representations for rigid solids: Theory, methods, and systems. *ACM Computing Surveys, 12*(4), 437–464.

Dassault Systèmes. (n.d.). *SOLIDWORKS help*. Retrieved [Month Day, 2026], from https://help.solidworks.com

## 8.3 Books

Planchard [1] was the main reference for SolidWorks features and workflow; Bhatt and Panchal [2] for orthographic projection and drawing conventions; Budynas and Nisbett [4] for general design background on shafts, bearings and fits; Boothroyd et al. [5] for manufacturing-oriented design; and Shah and Mäntylä [6] for the concepts of parametric and feature-based modelling.

## 8.4 Websites

The SOLIDWORKS Help documentation [8] was used to confirm the behaviour of features such as Flex, Rib, Shell and mates. Tinkercad [9] was used for the shape-based profile exercises. The assignment briefs [10] were the primary source for dimensions and steps.

## 8.5 Research Papers

The paper by Requicha [7] describes the representation of solids and is the foundation for the solid modelling used in CAD systems. Further papers read during or after the internship can be added here in IEEE format.

Table 10: Report formatting followed

| **Item** | **Format used** |
|---|---|
| Font | Times New Roman style serif, 12 pt |
| Headings | 16 pt bold (chapters), 14 pt bold (sections) |
| Line spacing | 1.5 in body text; 1.5 inch left margin, 1 inch on other sides |
| Page numbers | Bottom centre |
| Captions | Below figures, above tables |